



ADAMA SCIENCE AND TECHNOLOGY UNIVERSITY

SCHOOL OF ELECTRICAL ENGINEERING AND COMPUTING

Integrated Engineering Team Project

System Design Specification Document (SDS)

Project Name: SafeMesh: Offline-First Decentralized Communication Network

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1. Introduction

1.1 Purpose

This System Design Specification (SDS) outlines the technical architecture, hardware-software integration, components and implementation strategies for SafeMesh — an ultra-low-cost, solar-powered, offline-first decentralized mesh communication network. It provides a clear blueprint to guide development, testing, integration and field deployment.

1.2 Scope

The document focuses on the MVP for 2026 deployment, including ESP32 firmware, mesh networking protocol, gateway bridging, Flask backend and React monitoring dashboard.

Out of scope: Voice communication, native mobile apps, LoRa integration and commercial certification.

1.3 Audience

- Technical team (embedded systems, backend, frontend developers)
- Hardware engineers and field deployment officers
- Project supervisors and university evaluators
- Investors, donors, and innovation partners
- Government and NGO stakeholders in rural connectivity

1.4 References

- 🚩 SafeMesh SRS Version 1.0 (May 2026)
- 🚩 Espressif ESP-NOW Technical Reference Manual
- 🚩 MicroPython ESP32 Documentation
- 🚩 IEEE Std 1016-2009 – Software Design Descriptions
- 🚩 Ethiopian National Digital Transformation Strategy 2025

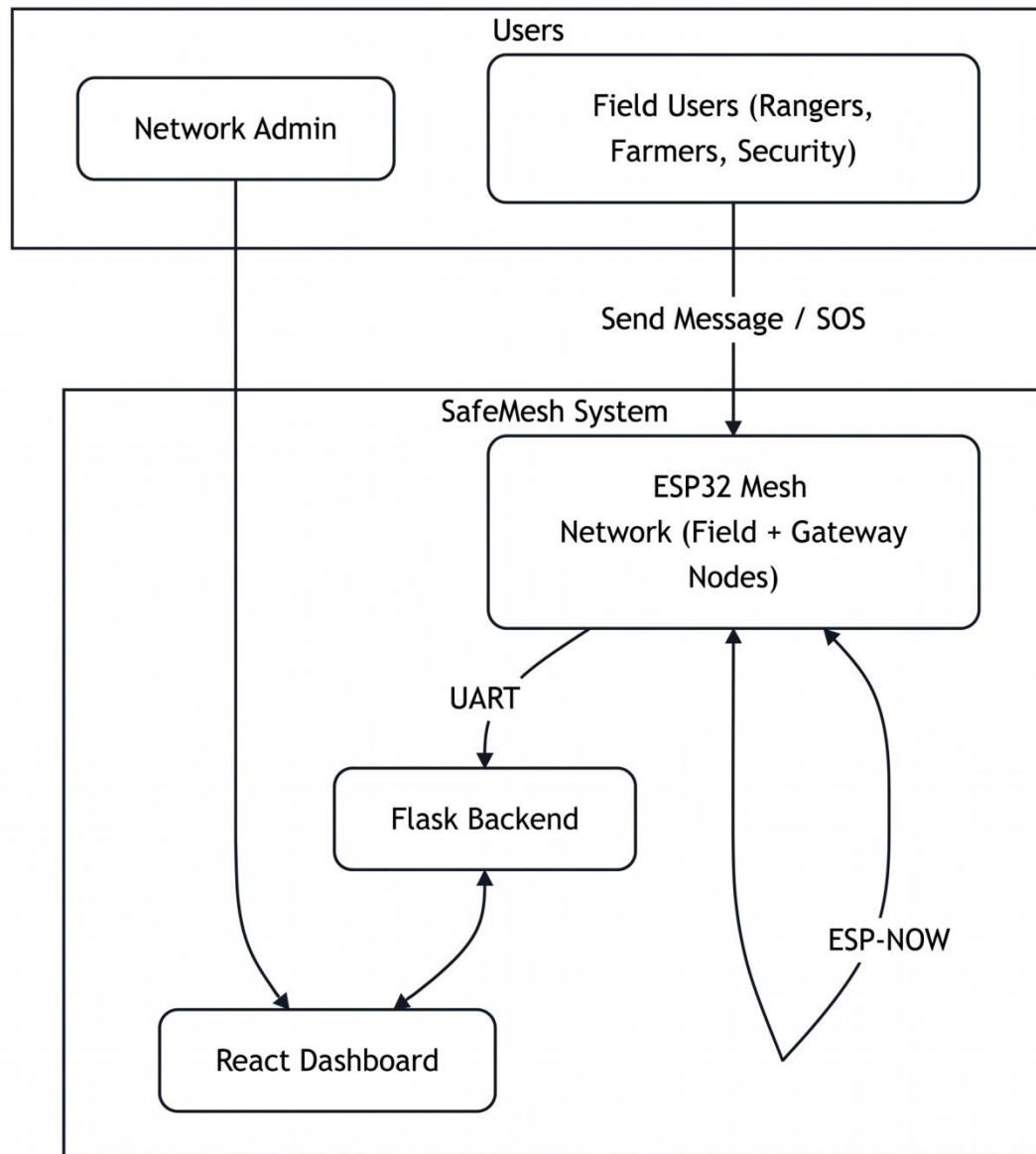
2. Design Principles

The SafeMesh architecture is guided by:

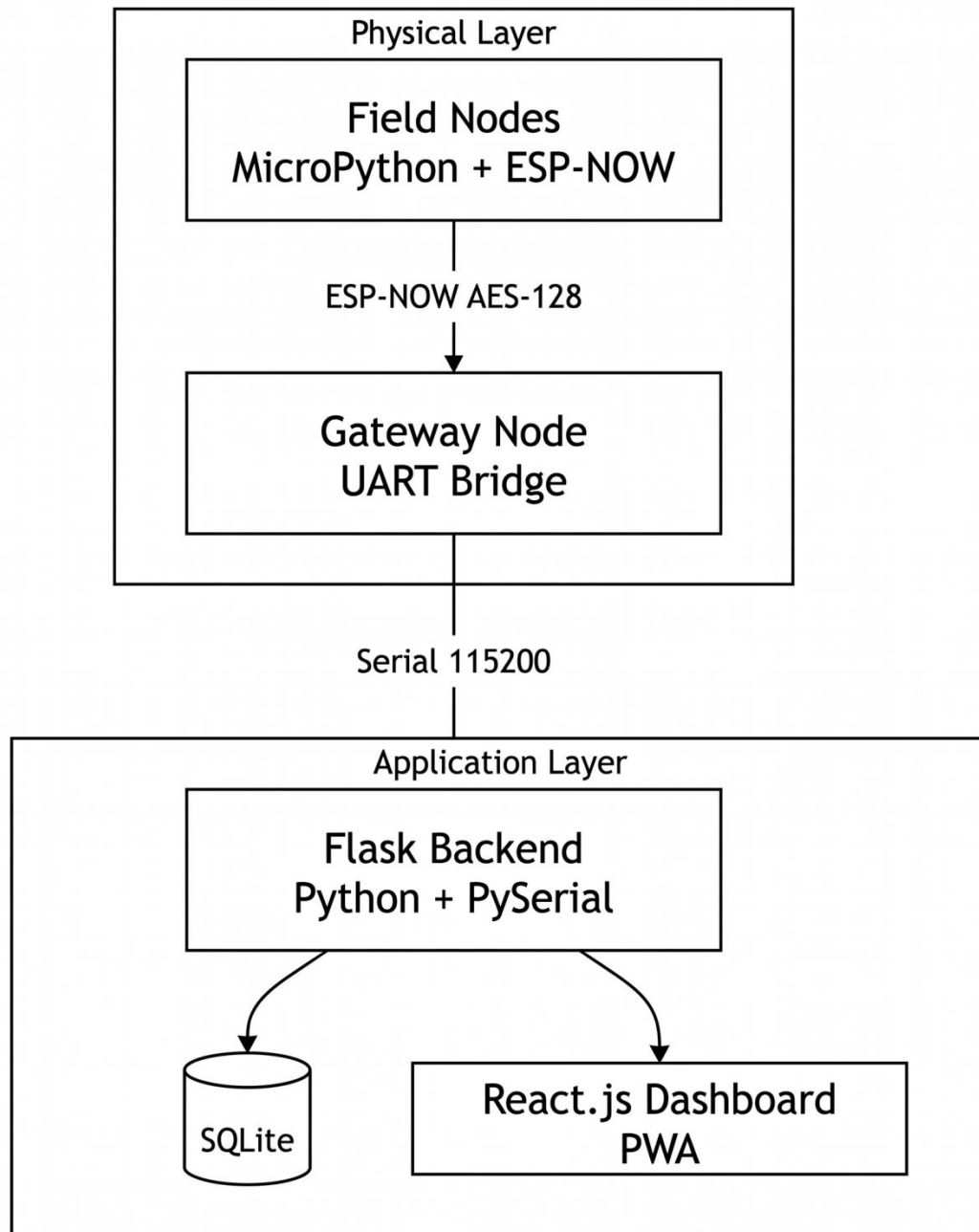
- ✚ **Ultra-Low Power Sustainability:** Aggressive deep-sleep cycles and solar-first design for weeks of operation.
- ✚ **Complete Offline-First Operation:** Full functionality without internet or cellular dependency.
- ✚ **Cost Efficiency:** Total node cost under \$8 USD using commodity hardware.
- ✚ **Security First:** AES-128 encryption with private network keys.
- ✚ **Reliability in Harsh Environments:** Controlled flooding with deduplication for robust delivery.
- ✚ **Hybrid Monitoring:** Independent radio mesh with optional centralized web dashboard.
- ✚ **Ease of Deployment:** Simple provisioning and modular firmware.

3. High-Level Architecture

3.1 System Context Diagram



3.2 Container Diagram



3.3 Deployment Architecture

Solar-powered ESP32 nodes are deployed across the target area. A single Gateway Node connects via USB/UART to a host computer (laptop or Raspberry Pi) running the Flask server and React dashboard for centralized monitoring and management.

4. Technology Stack

4.1 Firmware

- MicroPython v1.23+
- ESP-NOW library + ucryptolib (AES-128)

4.2 Backend

- Python 3.11 + Flask
- PySerial for gateway communication

4.3 Frontend

- React 18 + TypeScript + Tailwind CSS (PWA)
- Recharts & Mermaid for visualization

4.4 Hardware and Communication

- ESP32-WROOM-32, 18650 Li-ion + TP4056 + 5V solar panel
- Optional: NEO-6M GPS, SSD1306 OLED
- Protocol: ESP-NOW (2.4 GHz) + UART

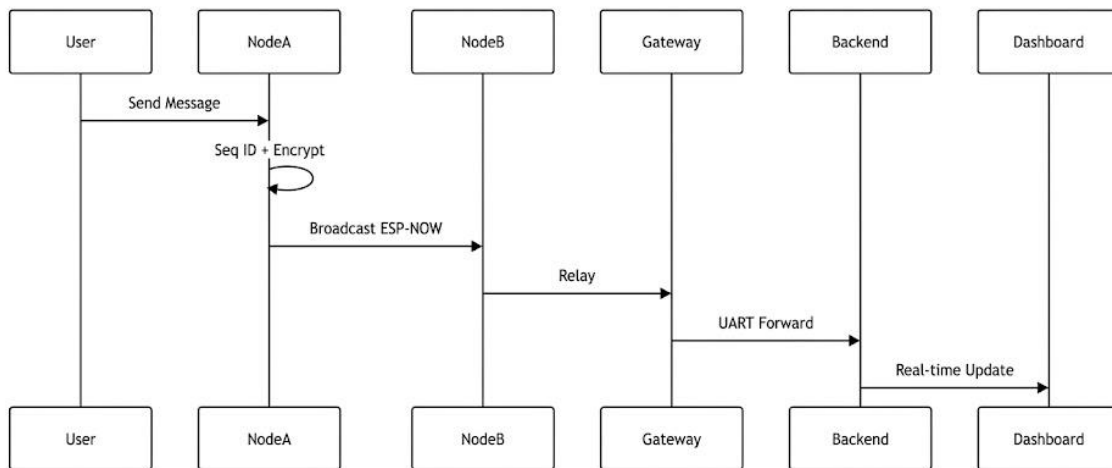
5. Detailed System Design

5.1 Component Breakdown

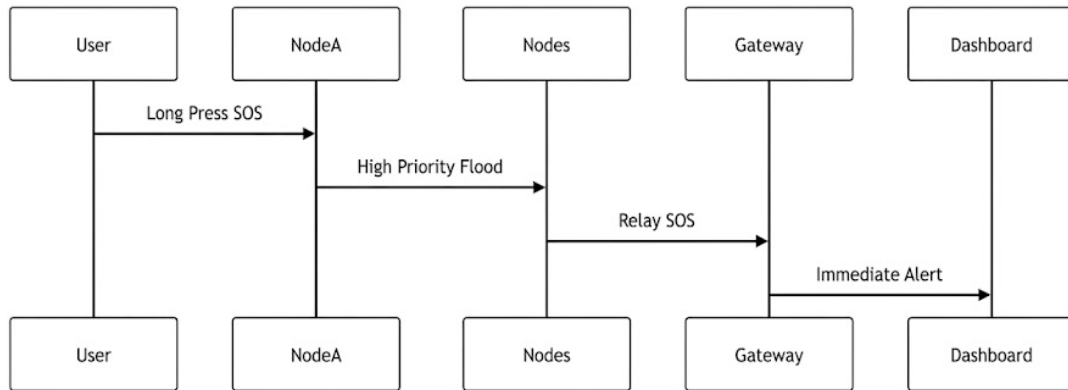
- ✚ **Node Firmware:** Message handling, relaying, encryption, power management
- ✚ **Gateway Bridge:** Serial packet forwarding
- ✚ **Backend Service:** Message ingestion, storage, and API
- ✚ **Dashboard UI:** Real-time visualization and alerts

5.2 Data Flow & Sequence Diagrams

Sequence Diagram: Normal Message Transmission



Sequence Diagram: SOS Emergency



5.3 Mesh Networking & Offline Mechanism

Controlled flooding with Sequence ID deduplication ensures reliable multi-hop delivery without complex routing tables.

5.4 Power Management & Solar Optimization

Duty cycling, deep sleep modes, and periodic telemetry optimize battery life for weeks on solar charging.

6. Data Model

6.1 Database Schema Overview

Lightweight relational schema optimized for message logging and node telemetry.

6.2 Key Entities

- ✚ nodes (node_id, mac, battery, last_seen)
- ✚ messages (seq_id, sender, content, timestamp, is_sos)
- ✚ telemetry (node_id, voltage, rssi, timestamp)
- ✚ network_keys (key_hash, created_at)

7. API Design

7.1 Endpoint Structure

Base: /api/v1 Examples: GET /messages, GET /nodes/status, POST /commands

7.2 OpenAPI Integration

Swagger UI auto-generated at /docs.

8. Security and Compliance

8.1 Authentication and Authorization

- Radio layer: AES-128 + Network Key
- Dashboard: JWT + Role-based access

8.2 Data Privacy & Encryption

All radio packets encrypted. Configurable message retention on gateway. Compliant with Ethiopian data protection regulations.

9. Performance and Scalability

9.1 Optimization Strategies

- Minimal packet size and compression
- Efficient deduplication and sleep cycles
- Caching on dashboard layer

9.2 Scaling Roadmap

- 🚧 Year 1: 10–50 nodes per mesh
- 🚧 Year 2: 200+ nodes with multiple gateways
- 🚧 Year 3: Multi-mesh federation support

10. Deployment and Operations

10.1 CI/CD Pipeline & Flashing

GitHub Actions for firmware build and dashboard deployment. Custom flashing tools for field use.

10.2 Monitoring and Logging

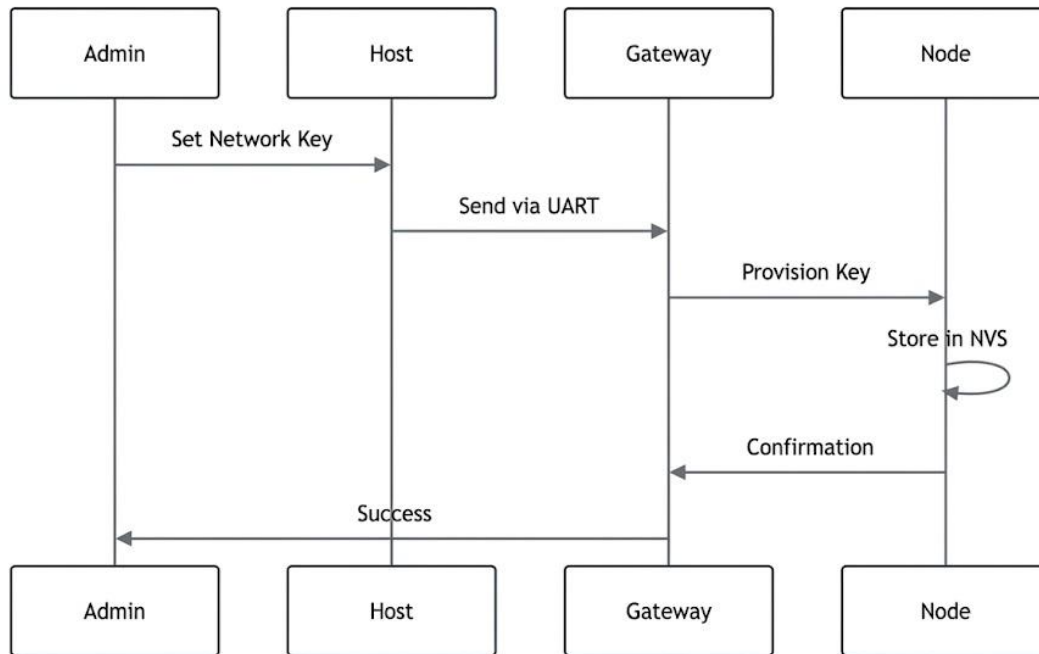
Real-time SOS alerts, node health dashboard and centralized logging.

11. Risk Assessment

- ✓ Radio interference: Mitigated by encryption and channel selection
- ✓ Power shortage: Solar optimization and battery telemetry
- ✓ Node failure: Self-healing mesh design
- ✓ Security breach: Private network keys and rotation support

12. Appendix – Additional Diagrams

12.1 Additional Sequence Diagrams



12.2 Bill of Materials & Cost Estimate

- Single Node: ≈ \$6–8 USD
- 10-Node Network: Under \$60 USD (including solar panels and enclosures)

13. Glossary

ESP-NOW – Espressif peer-to-peer wireless protocol **Controlled Flooding** – Reliable broadcast with Sequence ID deduplication **Sequence ID** – Unique message identifier for loop prevention **PWA** – Progressive Web App **AES-128** – 128-bit Advanced Encryption Standard

Conclusion

The SafeMesh System Design Specification outlines a complete, practical and cost-effective architecture for offline decentralized communication. Leveraging ESP32-based mesh networking with ESP-NOW protocol, controlled flooding, AES-128 encryption and solar-powered nodes, the system delivers reliable text messaging and emergency SOS capabilities in areas without cellular coverage.

This design successfully bridges low-power embedded systems with a modern web dashboard, ensuring scalability, security and sustainability. SafeMesh represents an innovative, high-impact solution that significantly reduces communication barriers in remote, rural and industrial environments at a fraction of traditional costs.

References

1. Espressif Systems. ESP-NOW User Guide (2025).
2. MicroPython ESP32 Documentation.
3. IEEE Std 1016-2009 – Software Design Descriptions.
4. GSMA Off-grid Connectivity Reports.
5. Ethiopian Digital Strategy 2025.